

Campus Sports Equipment Issue–Return System

A Digital-Credential Platform Replacing Physical ID-Card Collateral for Campus Sports Equipment Lending

Project Proposal for UCS503P

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1 Higher-order goal

This project aims to improve student welfare and the operational efficiency of campus sports infrastructure by removing the administrative friction built into the current equipment-lending process. While the immediate trigger is a single policy at one college — students must deposit their college ID card to borrow sports equipment — the underlying engineering objective is broader: design a deployable, reusable digital-credential and inventory-tracking system that can generalize to any campus asset-lending scenario (sports equipment today; library books or lab equipment tomorrow).

2 Time-to-value

- We will deliver meaningful increments quickly by prototyping the core issue–return workflow (digital token generation and staff-side scanning) and validating it with real students and sports-office staff within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations and CI-backed automation early, to keep release friction low.
- Success is defined by what can be delivered and measured in the first iteration — a working issue/return loop with timestamps — then improved based on user feedback.

3 Problem Statement

At present, students who want to borrow sports equipment (bats, rackets, footballs, etc.) from the college sports office must hand over their physical college ID card as collateral, which is returned only when the equipment is returned. In practice this creates real, recurring friction:

- Access barrier: students who forget their ID card cannot borrow equipment at all, and often end up borrowing a friend's card instead — which breaks accountability and misattributes usage.
- No visibility: there is no way to check equipment availability before walking to the sports office, so students frequently make a wasted trip and return empty-handed.

- Manual, paper-based tracking: issue and return records are kept in physical registers, making it hard to trace who has what, for how long, or in what condition.
- Weak enforcement: the only real penalty mechanism today is withholding the ID card itself, which is a blunt, all-or-nothing control rather than a graded, trackable consequence such as a fine.

Current Process	Proposed System
ID card held as physical collateral	Time-bound digital token; card stays with student
No visibility into availability before visiting	Live availability + slot reservation
Paper register, hard to trace history	Digital issue–return log with timestamps
Card withheld as sole penalty	Graded fines tied to no-dues clearance

Why this matters now (contextual relevance): A large college with many sports offerings has correspondingly heavy equipment turnover; even a modest reduction in wasted trips and untracked loans meaningfully improves both student experience and sports-office accountability.

4 Proposed Solution

4.1 Overview

Build a mobile-friendly web platform (Progressive Web App) that replaces the physical ID-card deposit with a time-bound, non-transferable digital credential, while borrowing the accountability model of a library issue–return system. Core components:

- Student portal: authenticate with college ID, browse live equipment availability by category and location, optionally reserve a time slot.
- Digital issue credential: a QR/token generated per request, bound to the student, the specific item, and a time slot — not a reusable general-purpose ID.
- Staff scanning app: a lightweight interface for sports-office staff to scan the token at issue and again at return, replacing the paper register.
- Fine engine: automatic fine calculation for late or damaged returns, feeding into the college's existing "no dues" clearance process at semester/degree end.
- Admin dashboard: inventory status, backlog, utilization, and most-borrowed items for the sports office.

4.2 Core workflow

- Student logs in with their college ID and browses live equipment availability, optionally reserving a slot.
- System generates a time-bound digital token for that student, item, and slot — the physical ID card is no longer required.
- Staff scan the token at the counter; the item is marked Issued, the due time starts, and the student keeps their ID card.
- On return, staff scan the same request again (or a return-side token) and record the item's condition; the item is marked Returned or Damaged.

- If the return is late or the item is damaged, a fine is automatically logged against the student's record and surfaced on both the student dashboard and the no-dues clearance system.

4.3 Operational constraints for an educational, tier-2/3 setting

- Keep the solution usable on low-to-mid range devices via a responsive Progressive Web App rather than a native app, avoiding install friction.
- Avoid dependence on advanced research algorithms; focus on workflow correctness, credential design, and accountability.
- Design for deployability using common web hosting and managed databases, with a hybrid (app + manual fallback) rollout during the pilot so the sports office is never fully blocked by a system outage.

4.4 Extension: Facility & Court Slot Booking

- The complex also has bookable facilities — badminton/tennis/basketball courts, the gym, the pool — not just loose equipment. Since the reservation engine in Section 4.2 already generalizes to "a student, an asset, a time slot," the same platform is extended to let students book a facility time slot directly, instead of only equipment:
- A facility (e.g. "Badminton Court 2") exposes bookable hourly slots, shown with live availability alongside equipment, using the same browse-and-reserve screen.
- Double-booking is prevented with a simple database-level uniqueness constraint/transaction on (facility, slot) at confirmation time — deliberately simpler than a distributed lock (e.g. Redis), which is unnecessary infrastructure at this scale and would work against the course's "avoid advanced/research-heavy infrastructure" guidance.
- No online payment or membership tier is introduced: access stays governed by the student's existing college ID rather than a paid subscription, keeping this an extension of the same credential system rather than a separate commercial booking product.
- This is scoped as a later-iteration, opt-in module (see Section 9.2) built on the existing token/reservation engine, not a parallel system — it is what makes the platform a genuine "sports complex" system rather than an equipment-only tool.

5 Solution Approach

This is an engineering course project, so the focus is on credential design, workflow correctness, and system reliability rather than novel algorithm research.

5.1 Digital credential design

- Each token is time-bound, non-transferable, and scoped to one student, one item, and one slot — unlike a general-purpose reusable ID.

- Unlike a single-scan canteen-style token, the credential is two-state (Issue scan, then Return scan), since equipment lending has two distinct events rather than one completed transaction.
- A manual staff-side override (ID lookup) is retained as a fallback for scan failures or network issues, so the desk is never blocked.

5.2 Web architecture

A typical 3-tier web architecture:

- Frontend: responsive PWA for students (browse, reserve, view token) and a simplified scanning view for staff.
- Backend API: authentication, inventory queries, token issuance/validation, status transitions, and fine calculation.
- Database: students, equipment items, batches/units, issue-return history, fines, and notifications.

5.3 CI/CD and fast delivery enabler

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion ... measurable and attributable

We define success using measurable metrics tied to the core workflow.

6.1 Primary evaluation metric

Time-to-Issue (TTI): the time between a student generating a digital token and receiving the physical equipment at the counter.

- Target: median TTI ≤ 5 minutes during the pilot window.
- Attribution: measured from database timestamps (token generation vs. staff issue-scan).

6.2 Secondary evaluation metrics

- Empty-handed rate: percentage of visits where a student could not obtain equipment due to unavailability, compared against a baseline collected before the pilot.
- Return-on-time rate: percentage of items returned before their due time.
- Fine-to-resolution rate: percentage of issued fines cleared before semester-end dues clearance.
- Staff processing time: time taken per issue/return transaction compared with the existing paper-register process.
- User clarity: student-reported clarity/satisfaction using a simple 1–5 feedback question.

6.3 Pilot validation plan

- Recruit 10–20 pilot users (students) and 2–3 sports-office staff for a two-week pilot run alongside the existing manual process.
- Compare metrics before/after within the iteration window by logging baseline estimates via short interviews and existing register data.

7 Scalability

The proposition should scale in both theoretical and practical senses.

Scalable (theoretically): the system should support growth in the number of equipment items, concurrent student requests, and dashboard usage by:

- decoupling components (frontend/backend),
- enabling pagination and indexing for common queries (availability lookups, student history),
- using stateless API design so multiple sports-office counters can operate concurrently.

Operationally deployable (practically): the system should run on common web hosting:

- managed database,
- containerized or platform-hosted deployment,
- CI/CD pipeline for staging/production.

Because the credential-and-inventory engine is generic (student, asset, slot, condition, fine), it is designed to extend beyond sports equipment to other campus asset-lending contexts — library books, lab equipment, or event gear — without a redesign, which is the project's primary scalability argument.

8 Engine availability heuristic

A mature set of "engines" (tooling/services) is available to implement this as a web-based project; indicative choices are listed below (final stack to be confirmed at design time):

- Frontend: React (or Next.js) as an installable Progressive Web App, for the student portal and staff scanning view.
- Backend API: Node.js/Express or FastAPI, layered as Route → Service → Repository.
- Database: PostgreSQL (managed instance) via an ORM (Prisma/SQLAlchemy) for schema migrations.
- Auth: JWT-based sessions, with an integration point for college SSO/ID verification where available.
- QR code generation (e.g. qrcode) and browser-based camera scanning (e.g. html5-qrcode) library.

- Notification channel for due-time reminders (in-app first via web push; email/SMS optional later).
- Test framework: Jest/PyTest for unit and integration tests.
- CI/CD: GitHub Actions runner, deploying containerized builds to a staging environment.

We will use these mature components to focus on workflow design, credential correctness, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

9.1 Initial deliverable

- Student login and live equipment browsing
- Digital token generation for a request
- Staff scanning app: issue-side scan and status update
- Basic logging and timestamps for TTI measurement
- CI: build + backend unit tests + linting
- CD to staging with a single command or pipeline job

9.2 Subsequent deliverables

- Return-side scan with condition capture (Returned / Damaged) and automatic fine calculation
- Live slot booking/reservation to prevent wasted trips
- Due-time reminder notifications (in-app, then email/SMS)
- No-dues integration or export for semester-end clearance
- Admin dashboard for utilization, backlog, and most-borrowed items
- Improved search/filtering and indexed queries for scale readiness

9.3 Indicative iteration timeline

Timeframe	Focus
Week 1–2	Setup, data model, student login, CI pipeline scaffolding
Week 3–4	Live availability browsing, digital token generation, staff issue-scan
Week 5–6	Return-scan, condition capture, automatic fine engine
Week 7–8	Slot booking/reservation, due-time notifications
Week 9–10	Admin dashboard, no-dues export, pilot with real users
Week 11–12	Feedback-driven fixes, scale-readiness pass, final evaluation

10 Risks and mitigations

- Authentication dependency: the system assumes access to a college SSO/ID-verification mechanism; if unavailable, fall back to admin-issued student credentials for the pilot.

- Staff adoption: paper registers are familiar and trusted; mitigate with a simple scanning UI, a hybrid rollout alongside the manual process, and a manual-entry fallback for scan failures.
- Damage/loss disputes: mandate a condition note (and optionally a photo) at both issue and return to avoid disagreements about pre-existing damage.
- No-dues integration access: the college's degree-clearance system may not expose an API; mitigate by exporting a fine list the sports office can reconcile manually during the pilot, with deeper integration explored later.
- Student adoption resistance: run a short awareness/onboarding push at pilot launch, since the workflow changes a habit students already know.

11 Summary

This project proposes a deployable web platform that removes the ID-card-as-collateral requirement from campus sports equipment lending, replacing it with a time-bound digital credential modelled on library issue–return accountability. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria — especially time-to-issue and empty-handed-visit reduction. It remains focused on engineering workflow, credential design, and scalability readiness (toward other campus asset-lending use cases) rather than research-driven novelty.